

Per-task gate threshold fitting

Qwen2.5-Omni-7B on OmniPro Online — the full record:
approach, results, figures, findings, and what none of it shows

1 September 2026

One-paragraph summary. A streaming assistant must decide *when* to speak. It decides by thresholding p_{hit} , a probability read off the frozen model’s logits. This study fitted that threshold *per task* over two grid passes on a frozen 135-video subset, then ran the complete 2,700-sample benchmark, which is now **finished: 2,700/2,700, gross time-F1 0.2265. The fit does not hold up, and the reason is measurable:** across all nine tasks the AUC of p_{hit} against event-adjacency is **0.394–0.526** — chance — so a threshold controls *how often* the system speaks and never *when*. Nine fitted thresholds do not beat one global threshold outside noise. The negative result is established four independent ways and is the contribution.

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1 What the system does and what is being fitted

A video plays. The model watches and listens as it arrives. At some moment it should *speak* — report a count, name a state change, alert on an event. The question is not *what* to say; it is **when**.

The deciding machinery is small. Once per tick the system asks the frozen model a yes/no question — “do we have enough information / has the event happened?” — and reads $P(\text{true})$ off the logits. Call it p_{hit} . If p_{hit} crosses a threshold, the system speaks.

Nine tasks share that mechanism, and the motivating guess was that they want *different* thresholds: a task that should alert once per video is not a task that should narrate continuously.

The benchmark. OmniPro Online: 9 tasks $\times$ 300 videos = **2,700** samples, 24.1 s–594.6 s, 65% requiring audio. **The metric.** `time_f1`: an emission counts if it falls within ± 3 s of a ground-truth trigger, matched greedily one-to-one. It needs no LLM judge, which is why every decision here is made on it.

The gate is three coupled knobs, not one: the threshold, the mode (`edge` = fire on a rising crossing, `level` = fire on every tick above), and the refractory (seconds of enforced silence after firing). **Only the threshold was refitted.** The other two are inherited unchanged from the vision-only predecessor. This is the study’s largest structural limitation and it recurs throughout.

2 Why the fit was necessary: the knob was disconnected

An audit of the shipped system found the threshold was not in the firing path at all. Under `gate_strategy="hysteresis"` the test that actually decided firing was `p_hit >= gate_high_thr`, a fixed global **0.5**; the per-task `hit_threshold` only decided whether an answer was *decoded*.

Across a completed 2,700-sample run — **581,403 ticks, 50,433 fires** — **not one fire occurred below** $p_{hit} = 0.500$. The four tasks configured at or below 0.5 were all silently running at 0.5. `task_gate_modes` and `task_refractory_s` were read **nowhere at all**.

The failure was **silent**: it produced a finished run and a number you could put in a table. It also means the “shipped” thresholds were never actually tested at their stated values — so this study’s baseline is itself a fiction that had to be re-measured.

2.1 A second defect, found while the fit was running

p_{hit} is high on the **very first tick** of every video: the model answers from the prompt before it has seen anything (≥ 0.05 for 98–100% of samples, median 0.34–0.64). Seeding the gate’s `prev_above` to **False** made that first tick a rising edge, so a long-refractory task spent its single allowed emission at $t \approx 1$ s against ground truth at 26–98 s.

The fix is definitional — an edge requires a predecessor, so `prev_above` starts **None**. Offline best time-F1 rose $0.0535 \rightarrow 0.0900$ (`snapshot_counting`) and $0.0636 \rightarrow 0.0751$ (`instant_event_alert`); short-refractory tasks moved ± 0.001 . **The size of the effect tracks the refractory**, which is the practical proof that the three knobs cannot honestly be fitted one at a time.

3 Approach

stage	what it does	cells	sample-evals	status
pass 1	wide grid, 10 thresholds $\times$ 9 tasks	94	1,410	complete
pass 2	refine 5 points around each pass-1 winner	45	675	complete
stage 3	the full benchmark at the fitted values	1	2,700	complete

The frozen subset. Passes 1–2 run on a fixed 5% all-audio subset — **15 videos per task, 135 total** — chosen once and never changed. Freezing it is what makes the two passes comparable to each other and to the bootstrap auditing them.

Protocol rules behind every number here:

- **Bootstrap resamples *videos***, never ticks or events. Ticks within a video are serially correlated through one shared KV cache; resampling them would manufacture confidence that is not there.

- **Paired, and it replays the selection rule.** Every threshold is scored on the *same* resampled video set, and the rule is re-run inside each draw rather than an argmax taken — so the audit measures the stability of *the procedure*, not of one number.
- **Re-selection is always divided by chance.** A 10-cell grid has a 10% chance rate; a 5-cell grid, 20%. Comparing raw rates across grid sizes is meaningless, and doing so would have made pass 2 look better than pass 1.
- **A 0.03 noise band.** The eval is not bit-reproducible; run-to-run variance on one fixed config was measured at F1 0.255 vs 0.051.
- **Content metrics are withheld, never estimated** when the judge is unreachable. A blank cell is a fact; a guessed cell is a lie that reaches a paper.

4 The fitted thresholds

task	shipped	fitted	gap vs rival	grid span	re-sel	chance
cumulative_counting	0.925	0.6167	−0.0017	0.048	48%	20%
dedup_counting	0.992	0.5500	+0.0243	0.055	70%	20%
event_narration	0.100	0.6500	+0.0037	0.023	74%	20%
explicit_target_grounding	0.500	0.7167	+0.0039	0.065	47%	20%
instant_event_alert	0.450	0.4500	+0.0645	0.065	63%	20%
realtime_state_monitor	0.800	0.4500	−0.0093	0.029	38%	20%
semantic_condition_alert	0.980	0.2167	+0.0091	0.066	42%	20%
sequential_step_instruction	0.010	0.1500	+0.0025	0.022	41%	20%
snapshot_counting	0.985	0.3833	+0.0667	0.133	56%	20%

The values move a long way from the shipped ones — but see §5 before treating any as a discovery. `instant_event_alert` did not move: it scored **0.000 at every threshold** in pass 1, so the rule correctly flagged it FLAT and kept the shipped value rather than picking a winner out of noise.

5 Findings

5.1 Finding 1 — p_{hit} does not rank event-adjacent ticks above quiet ones

This governs everything else. The ROC of per-tick p_{hit} against a ± 3 s positive label, now measured on **all nine tasks** with paired-over-video bootstrap CIs:

task	ticks	% positive	AUC [95% CI]
dedup_counting	4596	13.6	0.394 [0.35, 0.45] [†]
semantic_condition_alert	3323	10.9	0.431 [0.36, 0.53]
event_narration	4034	11.3	0.454 [0.41, 0.51]
sequential_step_instruction	3300	22.9	0.454 [0.40, 0.51]
realtime_state_monitor	2882	18.7	0.455 [0.39, 0.52]
cumulative_counting	2833	16.2	0.469 [0.43, 0.53]
explicit_target_grounding	2947	4.2	0.512 [0.40, 0.63]
instant_event_alert	4106	2.9	0.512 [0.44, 0.59]
snapshot_counting	3703	2.9	0.526 [0.43, 0.64]

[†] the only CI that excludes 0.5 — and it excludes it on the *wrong side*.

0.5 is chance, and eight of the nine CIs contain it. The obvious escapes were all checked and closed: not a clock offset (swept ± 10 s — no peak), not too tight a tolerance (± 10 s — unchanged), not a degenerate score (232–284 distinct values spanning 0.000–0.997).

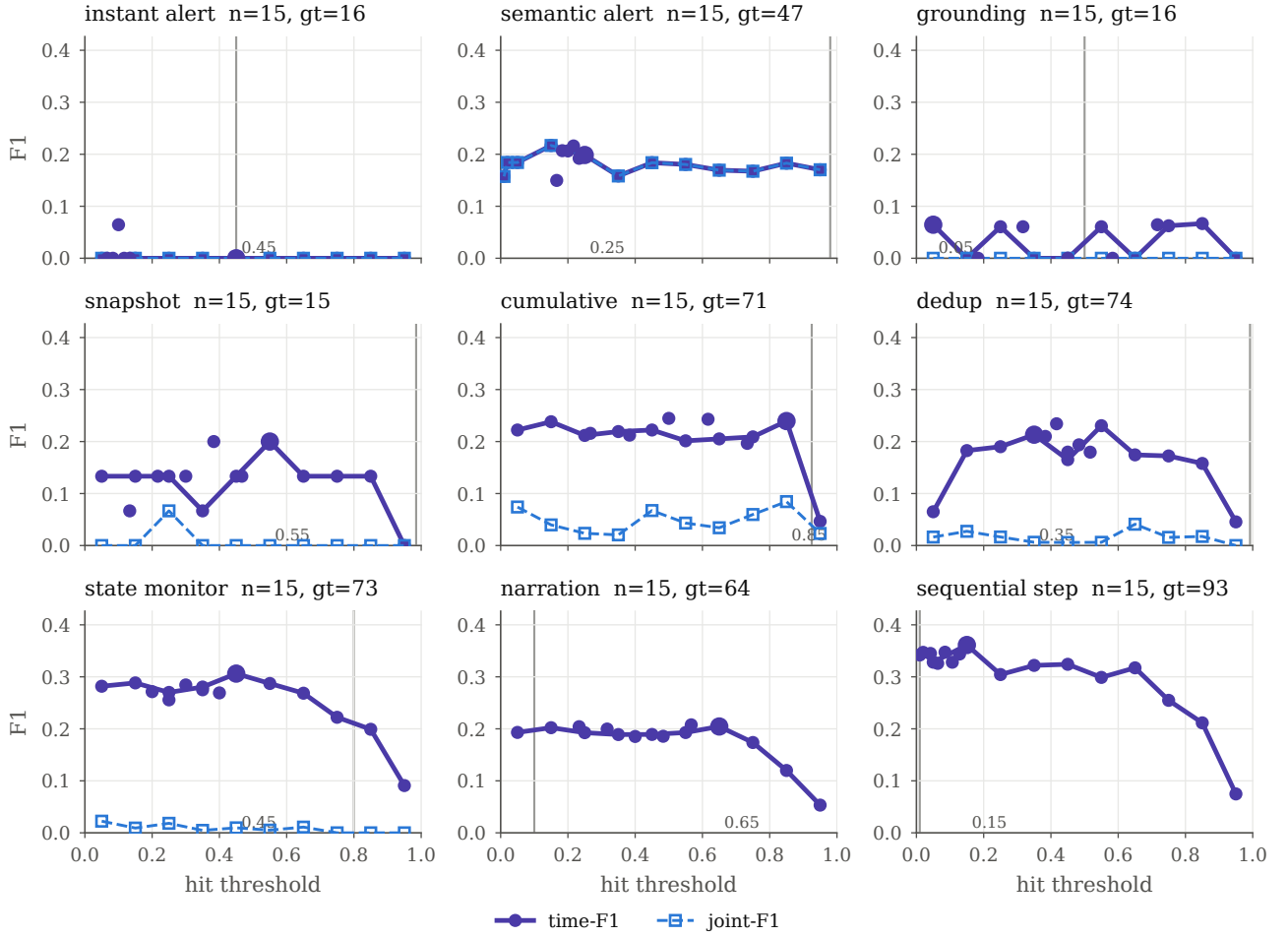


Figure 1: Pass 1, the wide grid. time-F1 against threshold for each task. The curves are flat across most of the range and collapse only at 0.95. A flat curve means the threshold is not selecting *moments* — it is selecting *volume*.

Two details worth pausing on. (i) **Eight of nine AUCs are below 0.5.** Pure noise would scatter symmetrically about 0.5; a median of 0.455 is a systematic tilt, consistent with the predecessor’s finding that p_{hit} correlates with wall-clock time ($|r|$ up to 0.93) rather than with content. (ii) **dedup_counting’s CI excludes 0.5 on the wrong side** — its p_{hit} is significantly *anti*-correlated with events. That is worse than uninformative: thresholding it actively selects the wrong moments.

5.2 Finding 2 — no task’s fitted threshold survives its own bootstrap

Pass 1 completed (1,410/1,410, 94/94 cells) and the rule returned a winner for all nine tasks. The paired bootstrap then found:

- **9 of 9 tasks have a $\Delta F1$ -against-best-rival CI containing zero.**
- Re-selection runs 25–61% against an 8–10% chance rate — `semantic_condition_alert` at just 6%, i.e. *below* chance.
- Two “fits” are artefacts: `instant_event_alert` scores 0.000 at every threshold, and `explicit_target_grounding`’s entire curve is one matched event or zero out of 16.

5.3 Finding 3 — the gate has one identifiable degree of freedom, not nine

All arms scored on the same videos at equal budget:

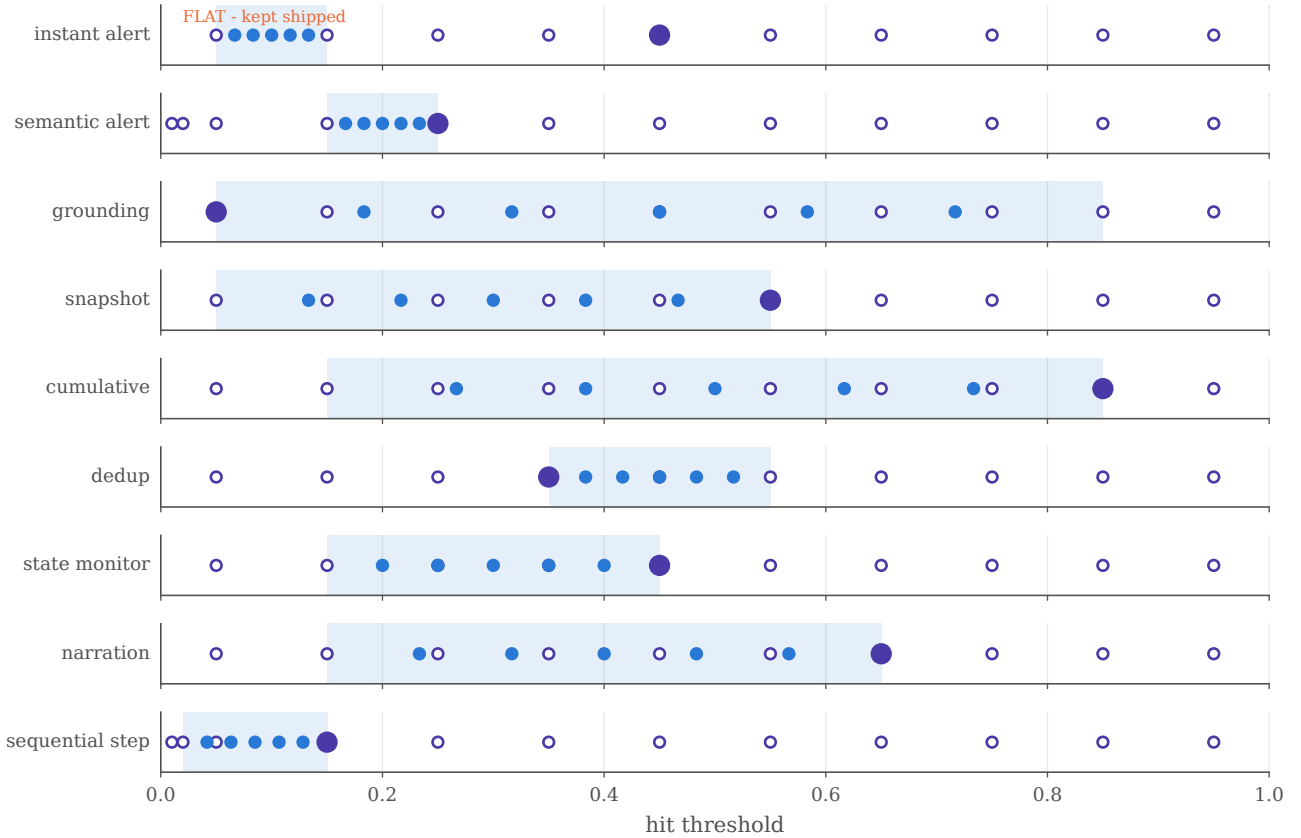


Figure 2: Pass 2, refined around each winner. The refined neighbourhoods span only 0.021–0.133 in F1, and on four tasks *every* cell falls inside the 0.03 noise band of the best. Refining did not resolve a winner; it confirmed there was no gradient to resolve.

arm	time-P	time-R	time-F1	vs. best single global
shipped_per_task	0.151	0.356	0.2115	−0.0354, CI [−0.055, −0.016]
global_0.5	0.142	0.565	0.2271	−0.0198, CI [−0.039, −0.003]
best_single_global (0.15)	0.155	0.601	0.2469	—
fitted_per_task	0.163	0.582	0.2544	+0.0075, CI [−0.011, +0.027]

Nine fitted thresholds beat one global threshold by **+0.0075 with a CI straddling zero** — *in sample*, where the nine have nine free parameters to its one. Yet the single global value is *solid*: re-running the sweep inside every bootstrap draw, **0.15 is re-selected in 94%** of them against 10% chance.

Pooled over 135 videos the operating point is identified; split nine ways over 15 videos each it is not. **That is a power problem, not a claim that the tasks are alike.** Note also that the *shipped* per-task thresholds are significantly *worse* than a flat 0.15 (−0.0354), and the damage is in recall (0.356 vs 0.601) — they were strangling the system. Replacing them is a real improvement even though the per-task *fit* is not.

5.4 Finding 4 — pass 2 confirmed a prediction registered before the data existed

Before pass 2 finished, a prediction was written down: the refined grid’s CIs would contain zero on *at least seven of nine* tasks. **The outcome was 9 of 9.**

The whole refined F1 span is 0.021–0.133, and on four tasks *every* cell sits within the noise band of the best. Two honesty notes are carried in the record: two rows report CI [+0.000, +0.200] whose lower bound is *exactly* zero and therefore does not exclude it; and pass 2’s re-selection of 38–74% beats pass 1’s 25–61% only until you divide by chance (20% vs 8–10%) — **as a multiple of chance, pass 2 is no better.**

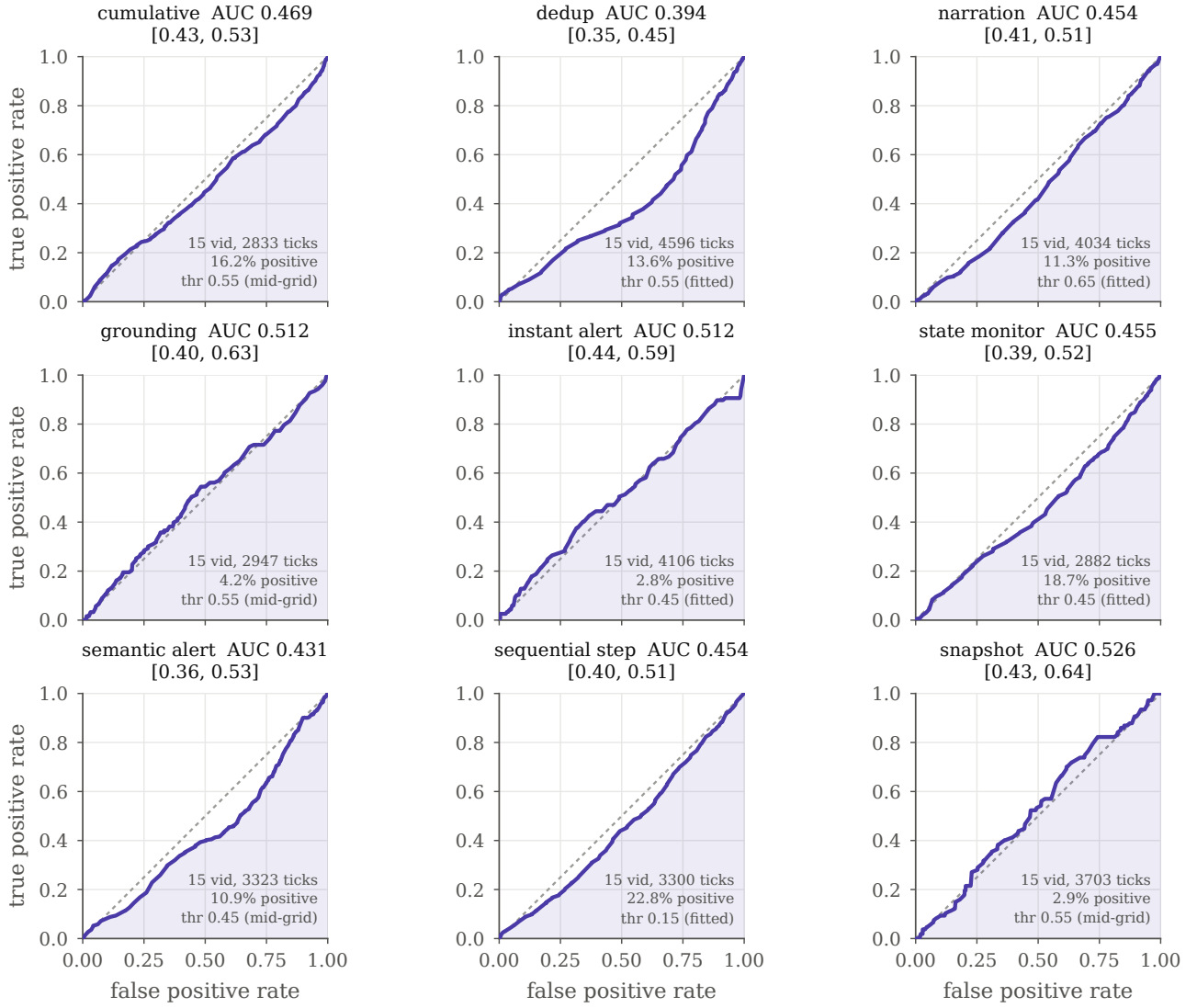


Figure 3: Finding 1, drawn. ROC of per-tick p_{hit} against $\pm 3s$ event-adjacency, one panel per task, with bootstrap CI, tick count, positive rate, and the threshold used. Every curve hugs the chance diagonal; `dedup_counting` bows *below* it. Panels state their own n so a 15-video panel cannot be mistaken for a 300-video one.

5.5 Finding 5 — the fit does not overfit

Stage 3’s 2,700 samples *contain* the 135 fitting videos, making the headline $\approx 5\%$ in-sample. Rescoring with those ids removed — free, the same predictions filtered — gives an in-sample bias of **+0.0003 gross** on the completed run (2,565 disjoint samples), and +0.0003 in both audio strata separately: two orders of magnitude inside the noise band. Whatever else is true of this fit, it generalises. The negative result is about *identifiability*, not overfitting.

6 Stage 3: the full benchmark, per task

All **2,700 of 2,700** banked — 2,700 unique ids, exactly 300 per task, **zero duplicates and zero torn records** across 76 generations and two lane reshapes, at a cost of **291.5 GPU-hours**. Judge offline, so content is withheld:

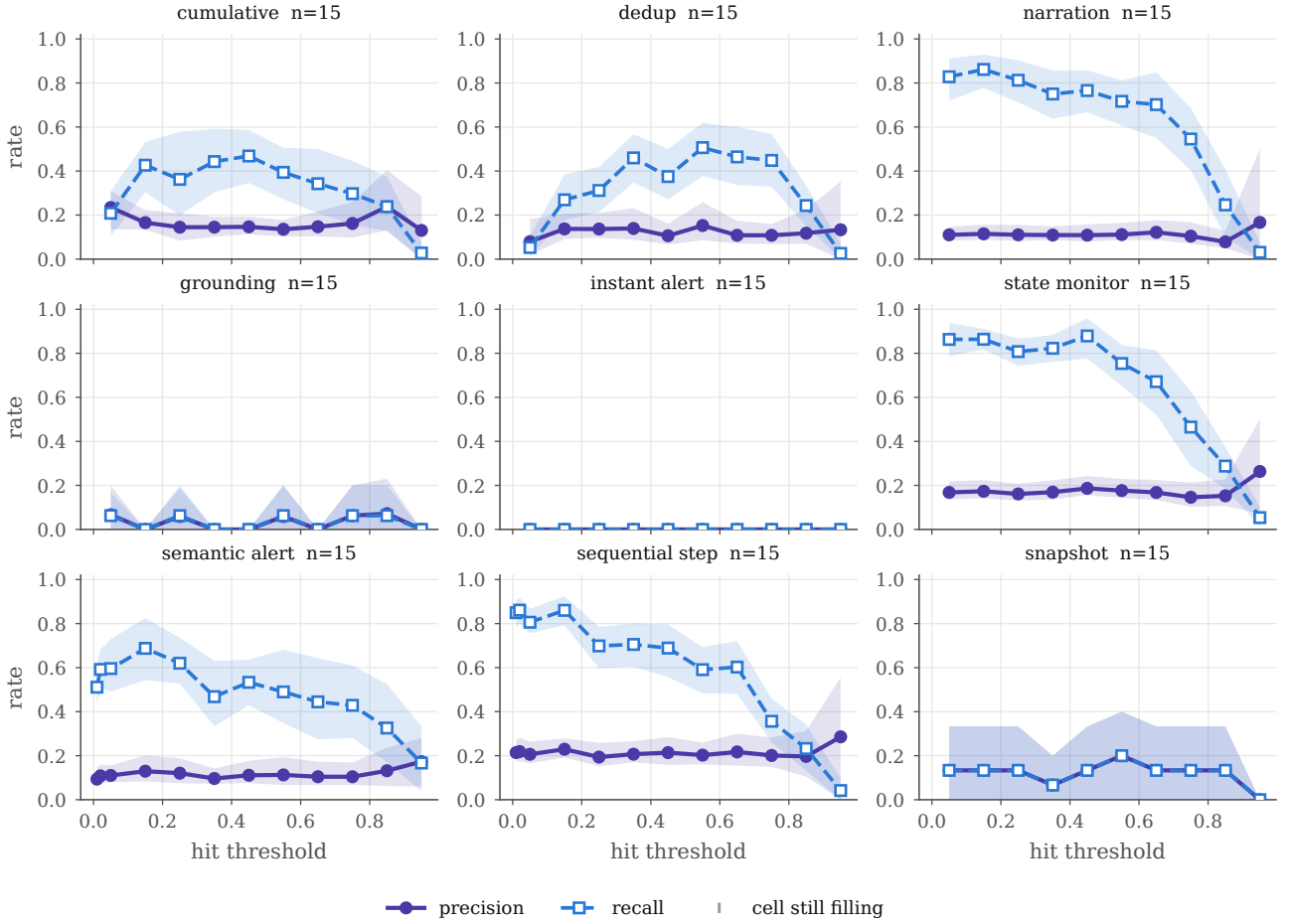


Figure 4: Precision is flat while recall falls. Sweeping the threshold moves recall and leaves precision alone — exactly what AUC ≈ 0.5 predicts, and arrived at through a completely separate code path from the ROC above.

task	n	n_{gt}	n_{emit}	emit/gt	t-P / t-R	time-F1
sequential_step_instruction	300	1709	5270	3.08	0.248 / 0.765	0.3748
cumulative_counting	300	1435	3704	2.58	0.152 / 0.393	0.2195
realtime_state_monitor	300	1340	8146	6.08	0.123 / 0.747	0.2110
event_narration	300	1322	6642	5.02	0.125 / 0.628	0.2084
dedup_counting	300	1234	5187	4.20	0.120 / 0.506	0.1944
semantic_condition_alert	300	984	4697	4.77	0.116 / 0.553	0.1915
explicit_target_grounding	300	332	332	1.00	0.099 / 0.099	0.0994
snapshot_counting	300	300	300	1.00	0.060 / 0.060	0.0600
instant_event_alert	300	395	300	0.76	0.067 / 0.051	0.0576
gross (micro)	2700	9051	34578	3.82	0.143 / 0.546	0.2265
macro over tasks						0.1796
audio required	1768	5288	19369	3.66	0.139 / 0.511	0.2190
audio not required	932	3763	15209	4.04	0.147 / 0.596	0.2363

6.1 Two failure regimes, not one

Regime A — over-emission (six tasks). 2.6–6.1 emissions per event. Precision, not recall, is what limits these. Gross precision is 0.143 against recall 0.546: the system finds the moments and buries them in false positives.

Regime B — one emission per video (three tasks). These carry 300s and 600s refractories inherited from the vision-only fit. `snapshot_counting` is the clean case: exactly 300 ground-truth events over 300 videos, exactly 300 emissions, and F1 **0.0600** — so the single emission lands within

Fig 7 Every interval crosses zero, in both passes

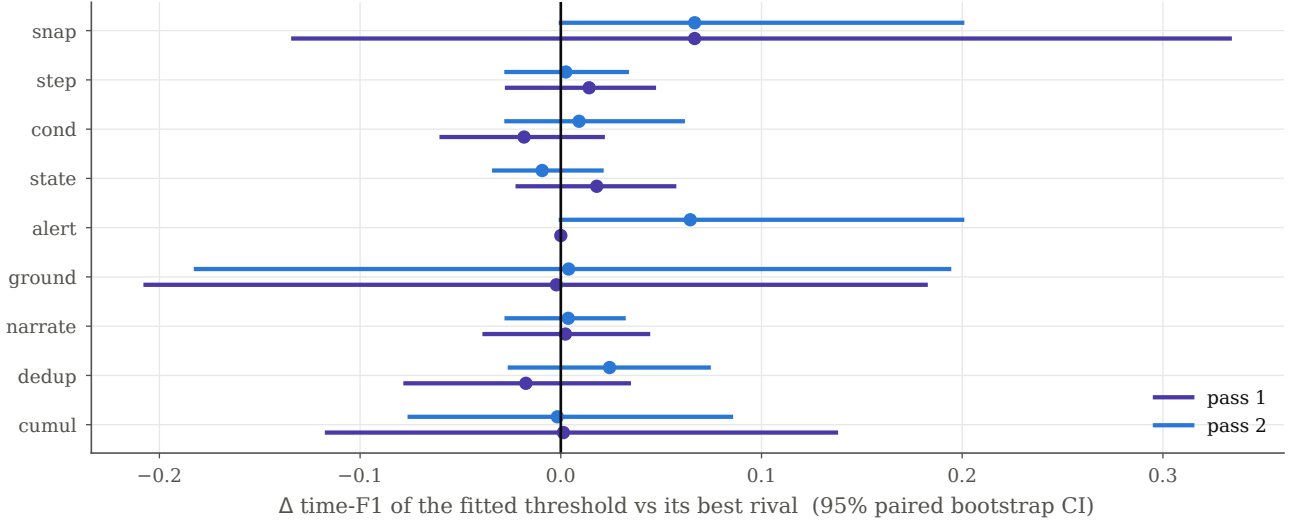


Figure 5: Findings 2 and 4 in one picture. Each task’s $\Delta\text{time-F1}$ of the fitted threshold against its best rival, with 95% paired-bootstrap CI, for both passes. **Every interval crosses zero.** Pass 2’s intervals (blue) are narrower than pass 1’s, but narrower is not the same as excluding zero. The two pass-2 intervals whose left end sits exactly on zero are the degenerate tasks matching 0–2 events out of 15–16; an inclusive bound of exactly zero does not exclude zero.

$\pm 3s$ exactly 6% of the time. The *count* is perfect; the *moment* is wrong. `explicit_target_grounding` is the same shape at 1.00 and 0.0994.

The gate is nearly exhausted, and the intuitive fix is backwards. With precision flat across the grid, $F_1 = 2pr/(p+r)$ rises monotonically as $r \rightarrow 1$ to a ceiling of $2p/(p+1) = \mathbf{0.250}$. We sit at **0.2265**, so the entire remaining headroom in the gate is **0.024** — inside the 0.03 noise band — and *cutting* the over-emission would **lower** F1, because precision will not rise to compensate.

6.2 Against the vision-only predecessor

	emits per event	micro time-F1	macro time-F1
system_3 (Qwen3-VL-8B, vision), $n=931$	1.48	0.2684	0.2565
omni (Qwen2.5-Omni-7B, A+V), $n=2700$	3.82	0.2265	0.1796

We win on **1 of 9** tasks on the completed run. Micro-averaged we trail by **0.042**, macro-averaged by **0.077** — both now outside the noise band, where the 95%-complete snapshot had the micro gap at a borderline 0.026. That movement is itself informative: **the long tail scores worse than the rest**, so every partial number in this study was optimistically biased and the final one is the only one to quote.

The two systems fail in opposite directions. The predecessor emits $1.48\times$ per event at precision 0.225 / recall 0.333; we emit $3.82\times$ at 0.143 / 0.546. It is conservative and accurate; we are talkative and vague. Our one win — `event_narration`, +0.039 — is a *level-mode* short-refractory task, exactly where talking more pays. Worth stating precisely: `realtime_state_monitor` is a *loss* by 0.0028, which is far inside the noise band, so the fair reading is **one clear win, one tie, seven losses**.

And where we lose most, it simply gets more attempts. On the three regime-B tasks the predecessor emits 2.85 / 3.07 / 1.33 times per event where our inherited refractories permit ≈ 1.00 . Inverting $F_1 = 2tp/(k+1)$ for one event and k emissions: it converts $\approx 53\%$ of `snapshot_counting` videos into a hit with 2.85 attempts ($\approx 24\%$ per attempt); we convert exactly 6% with one. So *both* factors are real — our per-emission timing is $\approx 4\times$ worse *and* we take a third as many attempts.

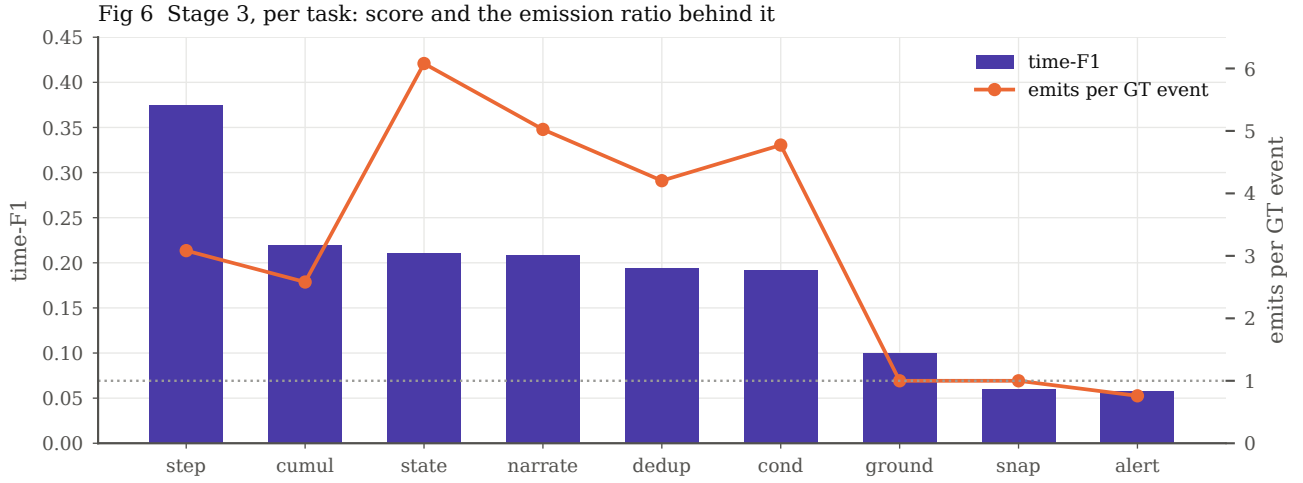


Figure 6: The result, and the mechanism beside it. Bars are per-task time-F1; the line is emissions per ground-truth event, with the dotted reference at 1.0. The two regimes are visible at a glance: six tasks emit $2.4\text{--}5.2\times$ too often, three emit about once per video.

7 Limitations — read before quoting any number

1. **Only one of three coupled knobs was refitted.** `mode` and `refractory_s` are inherited unchanged. The evidence above says they are coupled to the threshold, and §6.1 shows the cost concretely: the 300s/600s refractories permit ≈ 1 emission per video on the three worst tasks.
2. **There is no out-of-sample nine-versus-one comparison.** A control arm (flat global 0.15 over all 2,700) was built, preflighted, then **cancelled before launch**. The only fitted-vs-global number is the in-sample $+0.0075$, and it must always be cited as *in sample, $n=135$, nine free parameters against one*. Two predictions registered for that arm are **withdrawn, not resolved**.
3. **15 videos per cell is underpowered**, and that is the direct cause of Findings 2–4 rather than an incidental weakness.
4. **`content_acc` and `joint_f1` are withheld** throughout. No LLM judge has produced a verdict in any run of this fork: the workers `unset` the API keys deliberately, and the configured model (`gemini-3-flash`) 404s when tried. Seven of nine tasks are content-scored by deterministic parsers (`count`, `position`, `state`, or `time_only`); the two free-text tasks are unjudged, so the *pooled* figure cannot be formed. On the completed run the true joint-F1 lies between the lower bound **0.0349** and the timing ceiling **0.2265**, at 57% judge coverage. No comparison with published baselines (MiniCPM-o 4.5 at 0.209 *joint-F1*) is available — and they used a different judge on a different subset.
5. **The eval is not bit-reproducible.** Treat <0.03 as noise, always.
6. **The deepest one.** A separate diagnosis found the controller’s ICL schema *collapsing* on this backbone: a mean of **21.7 generated tokens against a 300-token budget**, with the self-pacing fields appearing in 20 ticks out of 78,828 and `answer` present on only 51%. The boolean this entire study thresholds is produced by that generation step. So Finding 1’s $\text{AUC} \approx 0.5$ is plausibly **an artefact of a broken upstream stage rather than a property of the task**. The findings remain correct *about the system as it ran*; what they predict about a repaired system is an open question.

8 Conclusions, and what to do next

Per-task threshold fitting did not work, and this study establishes *why*: not because the tasks are alike, but because the signal being thresholded carries no timing information, and because 15 videos

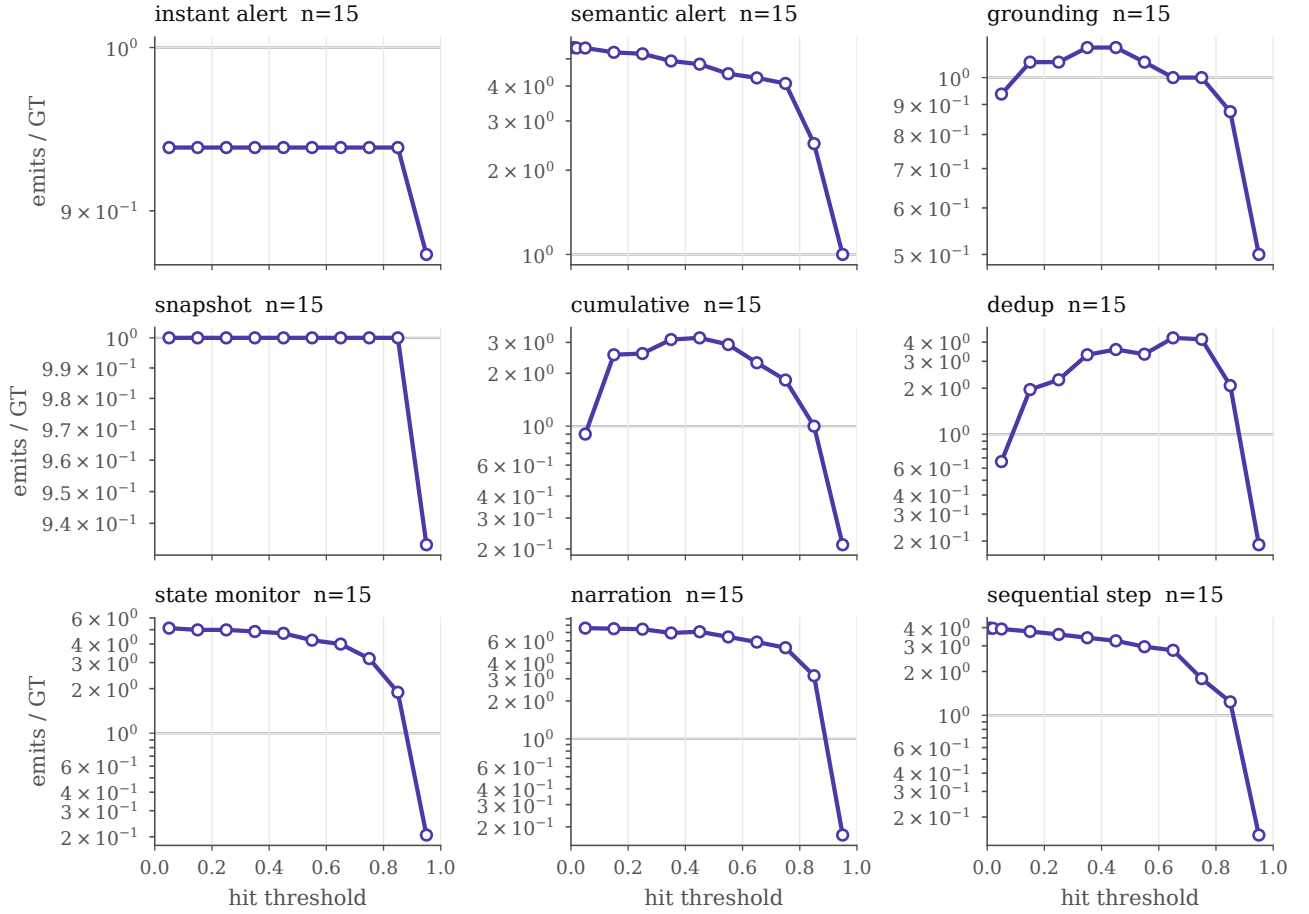


Figure 7: Emission calibration. How many times the system speaks against how many events exist. A well-calibrated gate would track the diagonal.

per task cannot identify nine parameters. A single global threshold (0.15) is the only operating point the data identifies, and it is identified well (94% re-selection against 10% chance).

Worth carrying forward:

- A negative result established by **four independent code paths** — ROC, flat precision curve, paired bootstrap, refined grid — with predictions registered before the data and *withdrawals recorded as visibly as confirmations*.
- The fit **generalises** (+0.0007 in-sample bias).
- The shipped thresholds are **measurably harmful** versus a flat 0.15.

The next move is not a better gate — with precision flat there is ≤ 0.03 left in it. Ranked by evidence:

1. **Use event_time_s, which is free.** The model already names an onset time on 41% of ticks and the gate discards it, timestamping emissions at the crossing tick instead. The data is already banked; testing costs no GPU.
2. **Make the controller finish its schema** (constrained decoding, or a smaller schema this backbone can complete).
3. **Refit mode and refractory**, which the head-to-head shows is worth more than the ceiling analysis alone suggested — it binds on the three worst tasks.
4. **Restore a judge.** Every content number is blocked behind it.

Fig 8 Head to head (time-F1 only; different subsets, see caption)

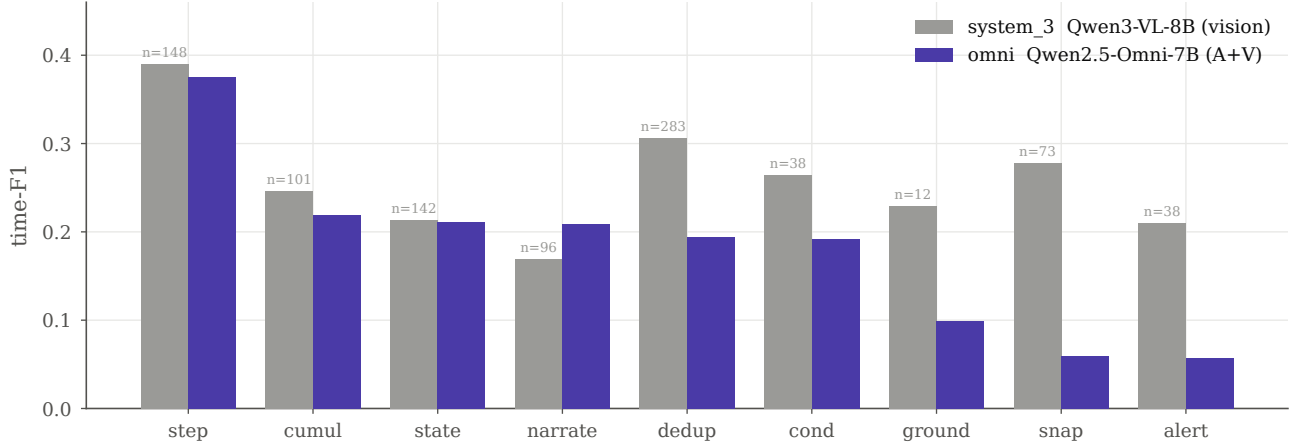


Figure 8: Head to head, time-F1 only. n is printed over each predecessor bar because it is uneven and small *exactly where that system wins most* (ground $n=12$, alert $n=38$) — those two comparisons carry the least weight in the chart, not the most. Different subsets, backbones and modalities: a signpost, not a controlled comparison.

Evidence. Every figure and number is recomputed from banked predictions and re-derivable without a GPU: AUC_TABLE.json and PERCEPTION_AUDIT.json (Finding 1), FIT_NOISE_AUDIT.json / _P2.json (2, 4), ABLATION.json (3), STAGE3_PARTIAL.json (§6), FINAL_THRESHOLDS.json, STAGE3_CONFIG_APPLIED.diff (what stage 3 actually loaded). Figures are built by lib/figs.py, lib/figs_auc.py, lib/figs_stage3.py. The full running record, including every deviation from the original spec and why, is THRESHOLD_FIT_RUNBOOK.md §2.1–2.10.